

Jharkhand University of Technology, Ranchi

B.Tech. 5th Semester Examination, 2023

Subject : Computer Graphics

Subject Code : CSC-503

Time Allowed : 3 Hours

Full Marks : 70

Candidates are required to give their answer in their own words as far as practicable.

All questions are of equal value.

Answer any five questions in which Question No. 1 is compulsory.

1. Choose the correct alternative:

- (i) Which of the following is a computer graphics type?
 - (a) Raster and Vector
 - (b) Raster and Scalar
 - (c) Scalar only
 - (d) All of these
- (ii) Which of the following plane is used for 2D transformation?
 - (a) 3D plane
 - (b) 2D plane
 - (c) One dimensional plane
 - (d) Four dimensional plane
- (iii) Good graphics programming avoids the use of _____ operations whenever possible.
 - (a) multiplication
 - (b) division
 - (c) floating point
 - (d) integer
- (iv) The simplest methods of filling a polygon is to examine _____ in the raster.
 - (a) 1 pixel
 - (b) 2 pixel
 - (c) 5 pixel
 - (d) Every pixel
- (v) The bounding box is the _____ that contains the polygon.
 - (a) smallest rectangle
 - (b) largest rectangle
 - (c) smallest triangle
 - (d) smallest square
- (vi) Which of the following a computer graphics curve?
 - (a) Bezies Curves
 - (b) Implicit Curves
 - (c) Explicit Curves
 - (d) All of these
- (vii) Which of the following is the purpose for using clipping in computer graphics?
 - (a) copying
 - (b) zooming
 - (c) adding graphics
 - (d) removing object and lines

$$-5 + 8$$

$$3 + 8 - 14$$

$$dy = 4$$

$$dx = 7$$

$$70 \text{ (next, } y_{k+1})$$

$$P_{k+1} = P_k + 2dy - 2dx$$

503

2. Write the Bresenham's line drawing algorithm and generate the points between the end points (2, 2) and (9, 6).

3. (a) Explain Raster scan display with diagram. 2, 1

(b) Describe DDA algorithm for circle with example.

4. What is the advantage of using homogeneous coordinates? Consider the square (0, 0), (0, 2), (2, 0), (2, 2). Perform the composite transformations of the square by using the following steps:

(a) Scale by using $s_x = 2$ and $s_y = 3$

(b) Rotate of 45 degree in the anticlockwise direction

(c) Translate using $t_x = 3$ and $t_y = 6$

5. (a) Explain the concept of Viewing pipeline for 2D transformation. 8

(b) Differentiate between parallel and perspective projection.

6. Briefly explain the steps involved in clipping a line using midpoint subdivision algorithm. A clipping window ABCD is specified as A(0, 0), B(40, 0), C(40, 40), D(0, 40). Using this algorithm find the visible portion, if any, of the line segment joining the points P(-10, 20) and Q(50, 10).